

SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_TX_Amarillo-Husband.Amarillo.Intl.AP.723630_TMYx.epw
- Building Name: SmallOffice
- Building Location: Amarillo, TX
- Total Floor Area: 511 m²

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof + window + door	Wall upgrade: Exterior walls were upgraded to R-13.0 with Fiberglass Batts insulation. Roof upgrade: The roof was upgraded to R-24.4 using Blown Fiberglass insulation. Window upgrade: • Upgraded to 3-pane glazing • Weatherstripping: silicone adhesive smoke gasket

- Glazing film: low-e film
- Window frame: wood window frame
- Caulking: acrylic
- Infiltration reduction: 30%

Door upgrade:

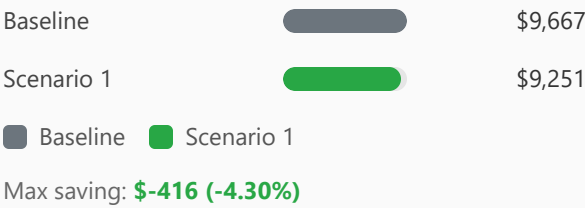
- Door type: wooden door
- Bottom seal: automatic door bottom
- Top/side seals: jamb weatherstrip
- Infiltration reduction: 30%

Annual Energy Analysis

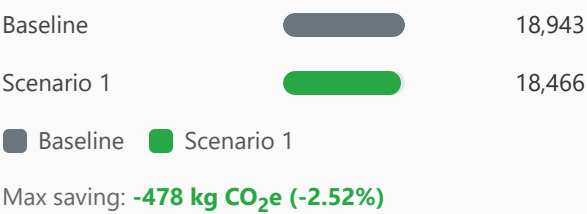
Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	195.48	190.41	5.07	2.59%

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison



Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr) Comparison



Min Retrofit Construction Cost

\$3,795

Scenario 1

Min Retrofit Embodied Carbon

19,345 kg CO₂e

Embodied carbon intensity (ECI): 38 kg CO₂e/m²
Scenario 1

Lowest Retrofit Cost Payback Period

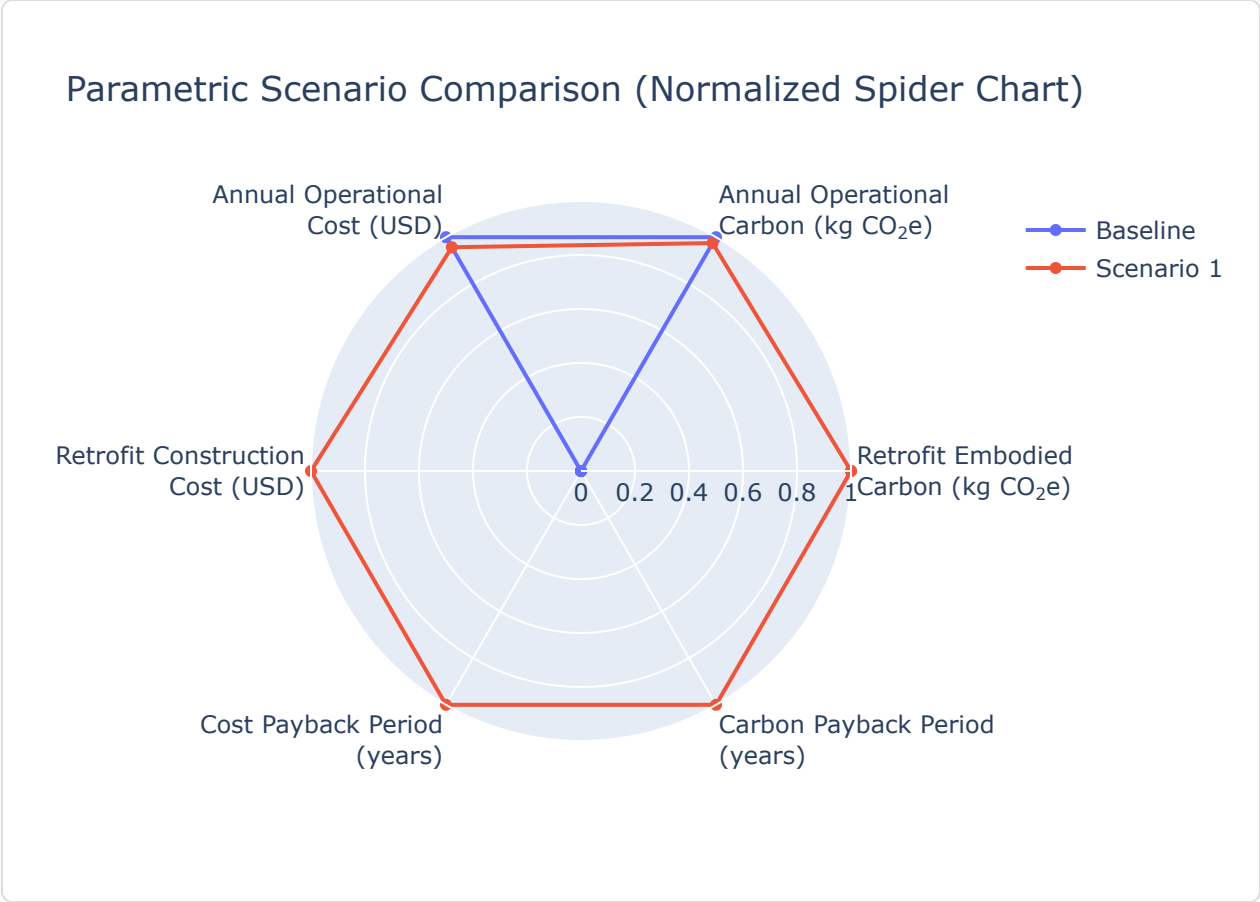
9.12 years

Lowest Retrofit Carbon Payback Period

41 years

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

Scenario 1 9.12 yrs

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr)).

Scenario 1 41 yrs

Material List

Material properties and consumption by renovation scenario. Missing values are shown

as N/A.

Scenario	Component	Material	Volume (m³)	Area (m²)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m³)
Scenario 1	Wall insulation	Fiberglass Batts	N/A	282	N/A	N/A	N/A	N/A
Scenario 1	Roof insulation	Blown Fiberglass	N/A	599	N/A	N/A	N/A	N/A
Scenario 1	Window weatherstripping	silicone adhesive smoke gasket	N/A	N/A	N/A	N/A	N/A	N/A
Scenario 1	Window caulking	acrylic	N/A	N/A	N/A	N/A	N/A	N/A
Scenario 1	Door bottom seal	automatic door bottom	N/A	N/A	N/A	N/A	N/A	N/A
Scenario 1	Door top/side seal	jamb weatherstripping	N/A	N/A	N/A	N/A	N/A	N/A

Result Summary

Scenario	Retrofit Embodied Carbon (kg CO ₂ e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO ₂ e)	Annual Operational Cost (USD)
Baseline	0.00	\$0.00	18,943	\$9,667
Scenario 1	19,345	\$3,795	18,466	\$9,251